DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Comparative Analysis of 1-bit Full Adders Using CMOS, PTL, TG and GDI

MINI PROJECT 2 | TRANSISTOR-LEVEL / ANALOG DESIGN

Prepared for: VLSI Design Project Submission

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Academic status: ECE, 3rd semester

Report status: Proposed design and verification plan. Numerical simulation results are not claimed until tools are run.

Abstract

This analog/transistor-level mini project compares four candidate 1-bit full-adder logic styles: static CMOS, pass-transistor logic (PTL), transmission gate (TG) and gate diffusion input (GDI). The objective is to establish a fair schematic-level simulation methodology for power, propagation delay, power-delay product and transistor count.

1. Problem statement

Implement functionally equivalent full adders using different transistor-level topologies and compare their characteristics under identical process, supply, load and input conditions.

2. Objectives

• Develop and verify four transistor-level full-adder schematics.

• Check sum and carry against the complete eight-row truth table.

• Compare average power, worst-case delay, PDP and transistor count.

• Explain trade-offs including degraded logic levels, sizing and output loading.

3. Literature review

Literature review scope: compare textbook CMOS inverter operation, transmission-gate switching, pass-transistor threshold loss, and published low-power VLSI cell architectures. Use verified papers from IEEE Xplore or institutional library to document technology node, supply voltage, testbench, and measurement methods before claiming one architecture is superior. No fabricated citations or performance figures are included.

4. Full-adder function and test setup

Sum = A XOR B XOR Cin; Cout = AB + ACin + BCin. Apply all eight input states plus input transitions. Use the same transistor models, VDD, temperature, input slew, output load and observation window for all styles.

5. Comparative simulation workflow

First build reference Boolean functionality. Next create each transistor-level schematic. Run DC/logic-level checks, transient simulation and power integration. Record device counts and propagation delays using consistent 50%-VDD crossings. Compute PDP = average power x worst-case propagation delay.

6. Experimental block diagram

A waveform generator applies the same A, B and Cin signals to each candidate full-adder cell. A common output load is attached to Sum and Cout; a measurement script compares output waveforms and integrated supply current.

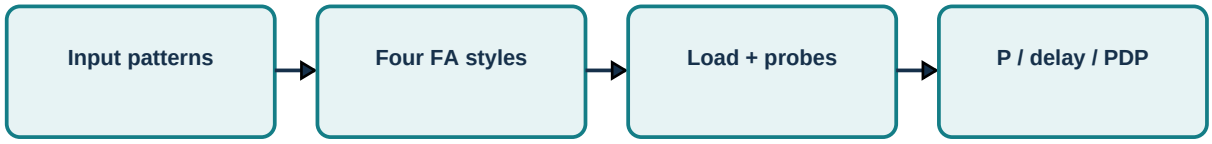

Figure 1. Fair, technology-matched comparative simulation workflow

7. Full-adder truth table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

8. Measurement table - to be filled after simulation

Logic style	Transistors	Avg. power	Worst delay	PDP
Static CMOS	TBD	TBD	TBD	TBD
PTL	TBD	TBD	TBD	TBD
TG	TBD	TBD	TBD	TBD
GDI	TBD	TBD	TBD	TBD

9. Practical design notes

PTL designs may require level restoration because an NMOS pass device passes a weak logic 1. Transmission gates use complementary controls and pass both logic levels more effectively. GDI structures depend on technology and body-terminal accessibility; a conventional bulk CMOS process may impose well/body restrictions. Raw transistor count is not sufficient to establish layout area.

10. Tools and reproducibility

Use LTspice/ngspice for preliminary circuits, or Cadence Virtuoso/Spectre if a valid PDK is available. State MOSFET model source, supply, load, corner, temperature, input sequence, transistor dimensions and timing definition in the final report.

11. Expected discussion

The comparison may reveal a power/area/speed trade-off, but the best style cannot be determined without a technology-matched simulation. No measured performance is asserted in this proposal.

12. Conclusion

The project links Boolean arithmetic with realistic MOS-level constraints and establishes a reproducible method for comparing logic styles.

13. References and further reading

N. H. E. Weste and D. Harris, CMOS VLSI Design; J. M. Rabaey, A. Chandrakasan and B. Nikolic, Digital Integrated Circuits. Add independently verified paper citations in the final literature review.